# Safestep

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https://github.com/rebeccayan9-dot/515FinalGroup

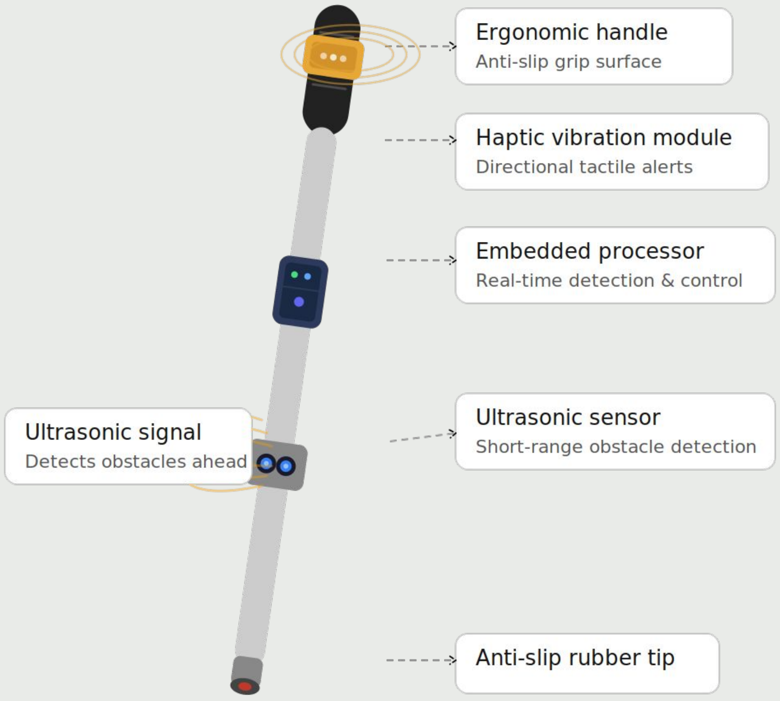
# Problem Statement

- Visually impaired pedestrians often struggle to navigate everyday environments safely. Traditional white canes are effective for detecting obstacles on the ground, but they can miss objects within the walking path.
- Existing assistive technologies try to address this gap, but many rely on external infrastructure or fail to provide clear and timely directional feedback. As a result, users often receive information that is delayed or difficult to interpret.
- This highlights the need for a lightweight and self contained system that can **detect obstacles, predict potential collisions, and communicate direction through intuitive vibration feedback.**

# Solution

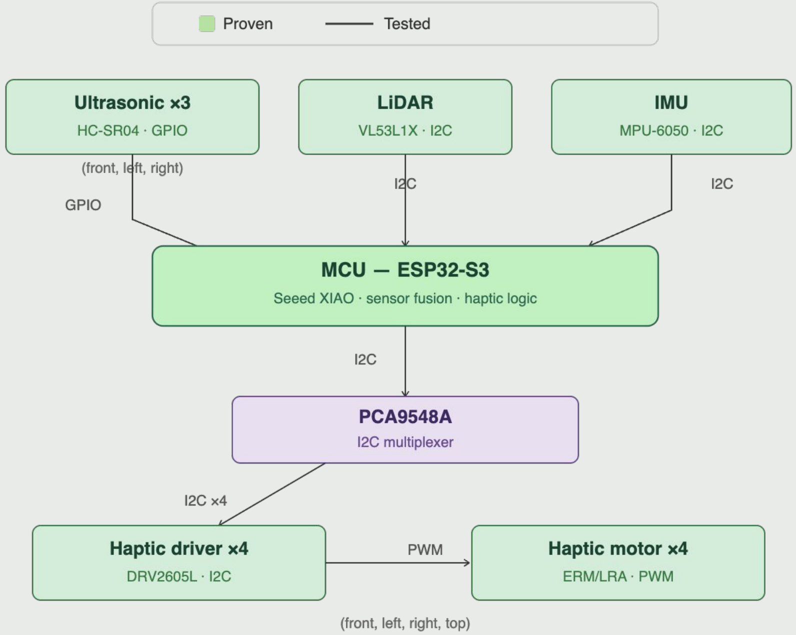
This system helps visually impaired users **detect nearby obstacles and understand their direction through vibration feedback.** It turns spatial information into clear, immediate cues, allowing users to identify whether obstacles are on the left, right, or directly ahead, and react quickly for safer navigation.

The system uses onboard sensing, embedded processing, and haptic output to provide timely guidance. It focuses on short range obstacle detection and directional alerts, rather than replacing a full mobility aid or vision based navigation system.




Smart navigation cane — system overview

# Update Architecture




**System architecture — Milestone 2**

Status per tracker Week 3

# Validation

1. data collection, model selection & training
2. Transfer&update system to PCB
3. TPU test print on handle prototype

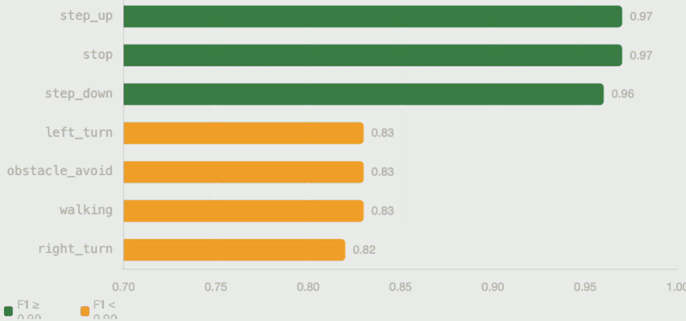
# 1- data collection, model selection & training

**Validation objective:** Verify that sensor data collected from the robot platform is sufficient and well-labeled to train a real-time behavior classifier that runs on the embedded target (ESP32-S3), and that the chosen model achieves acceptable accuracy across all defined behavior classes.

**Setup & method:** Multiple participants each performed all 7 behavior classes while navigating with the cane in a sight-free manner; sensor streams were logged over serial and saved as per-session labeled CSV files.Each row captures: timestamp (s), LiDAR distance (cm), front/left/right ultrasonic distances (cm), 3-axis accelerometer (g), and 3-axis gyroscope (dps), plus a behavior label.Data was split 80/20 for train/test, and a Random Forest classifier (n_estimators=50, max_depth=10) was evaluated with 5-fold cross-validation.

**Evidence:** Total 96 CSV files (~8,788 labeled samples) were collected across 4 sessions spanning April 22, April 28, April 29, and May 8, 2026, covering 7 behavior classes: step_up21,step_down17,walking17,obstacle_avoid14,left_turn11,right_turn10,stop6

**Quantitative results:**The model achieved 91% accuracy on the held-out test set (80/20 stratified split, 837 test windows) and 79.3% ± 10.3% on 5-fold cross-validation. The gap and high CV variance reflect class imbalance across recording sessions rather than overfitting.
On the deployment side, the serialized model is ~1.9 MB (joblib pickle). Python inference on a single 5-sample window averages ~14.6 ms, well under the 667 ms window stride (STEP=2 at ~3 Hz). The planned micromlgen C++ export reduces the on-device footprint further; the ESP32-S3's 512 KB SRAM and 240 MHz Xtensa core are sufficient for a depth-10, 50-tree forest running in real time.




Predicted

| | step_up | stop | step_dn | left_trn | obs_avd | walking | right_trn |
|---|---|---|---|---|---|---|---|
| step_up | 174 | | 5 | | | | |
| stop | | 51 | | | | | |
| step_down | 3 | | 142 | | | | |
| left_turn | | | | 81 | | 7 | 5 |
| obs_avoid | | | | 5 | 96 | 10 | 5 |
| walking | | | | 10 | | 129 | |
| right_turn | | | | 3 | | 11 | 69 |

Correct (diagonal) Key confusion (≥

# 2- Transfer&update system to PCB

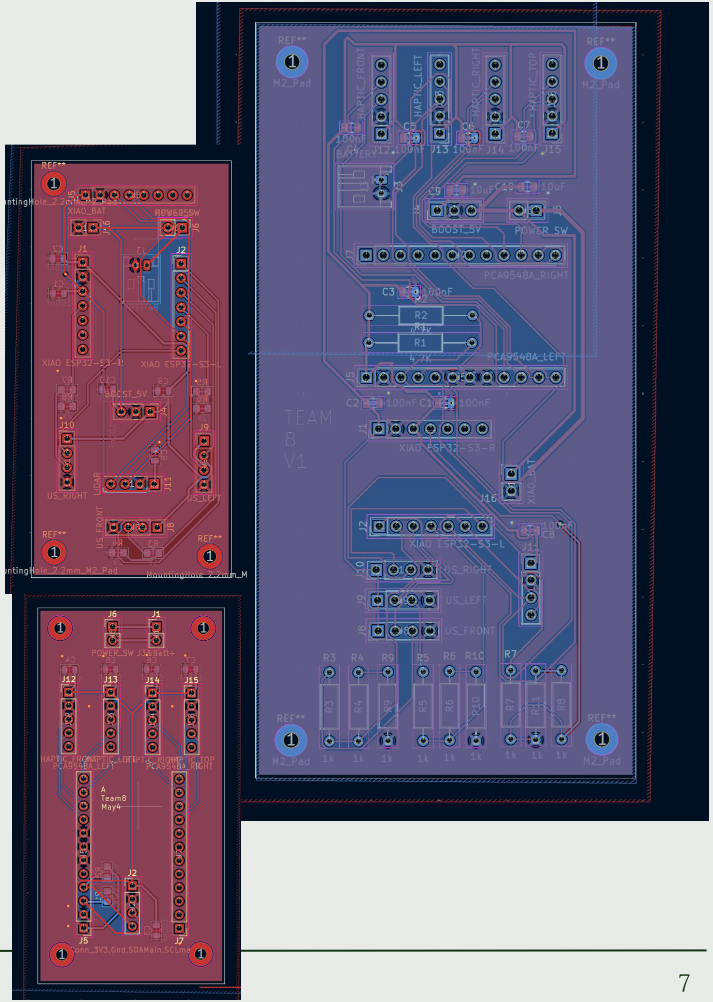
**Validation objective:** Confirm that breadboard-proven sensor and haptic circuits transfer correctly onto a custom PCB without functional regression. Verify newly introduced subsystems.

**Setup & method:** 1. Captured working breadboard circuit as a full schematic; added battery connector, power switch, 3.3 V LDO regulator, and PCA9548A I2C mux.2. Produced two PCB layout variants: Single-board with all components on one unified PCB & Split-board with main MCU/sensor board + narrower haptic driver/motor board, connected via flex header. 3. Go through In group review,DRC and ERC on both variants.

**Evidence:** screenshots, PCB run files.

**Quantitative results:**Components: all M1 proven blocks + 4 new (battery, switch, LDO, PCA9548A).Net count vs. breadboard — 5V, 3V3, Battery, GND, SDA main, SCL main.

**Conclusion:** PCB design submitted PCB run 3 awaiting for arrival and hand-soldering.



# 3- TPU test print on handle prototype

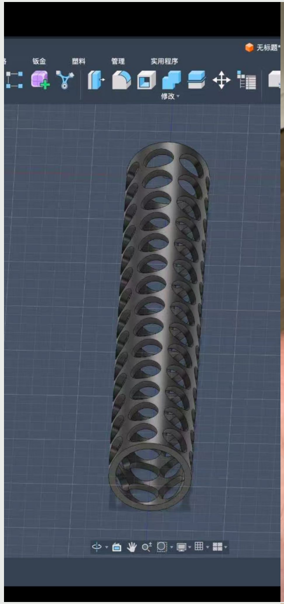
**Validation objective:** Verify that a TPU-printed handle prototype can house the haptic motor array at the correct size and spacing, assess TPU shore hardness and flexibility for handheld comfort, confirm FDM printability at target wall thickness, and validate that motor placement aligns with the positional requirements of a handheld blind-navigation device.

**Setup & method:** Modeled a first-revision handle shell sized around the 4× haptic motor pockets. Printed two TPU variants on FDM, with developed 3D model and material with different property.

**Evidence:** pic(upper-successful trial, blow-failed trial)

**Quantitative results:** Thinner-wall printer #1: print failure. Thicker-wall printer #2: print success motors fit with expected clearance.

**Conclusion:** First handle prototype confirms material physical property, doability and motor pocket geometry.




# Budge update

| Order Date | Order ID | Account Group | Item Net Total | Order Quantity | Currency | Order Subtotal | Order Shipping & | Order Promotion | Order Tax | Order Net Total |
|---|---|---|---|---|---|---|---|---|---|---|
| 05/08/2026 | 114-5472346-90 | Team 8 | 9.92 | 1 | USD | 8.99 | 0 | 0 | 0.93 | 9.92 |
| 04/26/2026 | 114-0299454-43 | Team 8 | 36.61 | 1 | USD | 33.19 | 0 | 0 | 3.42 | 36.61 |
| 04/21/2026 | 114-7331051-96 | Team 8 | 22.82 | 1 | USD | 20.69 | 0 | 0 | 2.13 | 22.82 |
| 04/21/2026 | 114-3694501-10 | Team 8 | 42.99 | 3 | USD | 38.97 | 0 | 0 | 4.02 | 42.99 |
| 04/15/2026 | 114-9467053-32 | Team 8 | 10.37 | 1 | USD | 9.89 | 0 | -0.49 | 0.97 | 10.37 |
| 04/14/2026 | 114-1914032-81 | Team 8 | 14.66 | 1 | USD | 13.29 | 0 | 0 | 1.37 | 14.66 |
| 04/13/2026 | 114-9170887-84 | Team 8 | 18.36 | 2 | USD | 29.64 | 0 | 0 | 3.05 | 32.69 |
| 04/13/2026 | 114-9170887-84 | Team 8 | 14.33 | 2 | USD | 29.64 | 0 | 0 | 3.05 | 32.69 |
| 04/13/2026 | 114-3326167-47 | Team 8 | 11.02 | 1 | USD | 9.99 | 0 | 0 | 1.03 | 11.02 |
| 04/07/2026 | 114-1902189-62 | Team 8 | 22.49 | 1 | USD | 20.39 | 0 | 0 | 2.1 | 22.49 |
| | | | | | | | | | | |
| | | | | | | | | | Total spend: | 236.26 |

# Thank You!